

Executive Summary Report

IEEE-CIS Fraud Detection, Advanced Risk Segmentation & Explainable AI Solution

Project Domain: Financial Crime Analytics / Fraud Operations **Date Generated:** May 25, 2026

Analytics Framework: LightGBM, XGBoost, SHAP, Optuna **Status:** Final Production Release Asset Framework

1. Executive Project Overview

This report serves as the formal comprehensive documentation summary for the end-to-end Machine Learning ecosystem developed to detect card payment fraud using the benchmark IEEE-CIS fraudulent transactions dataset. Financial fraud imposes billions of dollars in annual collateral losses. The core objective of this engineering pipeline was to navigate massive data sparsity, extreme class imbalance (only 3.50% fraudulent records base-rate), and construct a production-ready system consisting of robust feature transformations, optimal binary classifiers, distinct threat risk segmentation matrices, and a highly responsive operational Streamlit supervisor control center dashboard.

2. Pipeline Pipeline & Modeling Methodology

The lifecycle of data through the engineered pipeline underwent strict technical stages to eliminate data leakage and safely down-sample high variance features:

- **Data Merging & Feature De-Duplication:** Large-scale transactional vectors (590,540 records × 394 columns) were matched against unique client digital identification nodes (144,233 records × 41 columns) on cross-identity indices to build a single composite dataset.
- **Sparsity Filter & Imputation:** All missing metrics across variables were quantified. Columns exceeding a 50% missing-value threshold (214 columns) were dropped to guard model convergence. Remaining numeric gaps were imputed using median statistical aggregates, and categorical features were filled with modes.
- **Feature Engineering:** Three powerful predictive features were injected: *AmtToMeanRatio* (quantifying contextual transactional spikes), *HourOfDay* (capturing high-velocity localized time cycles extracted from elapsed delta seconds), and *LogTransactionAmt* (normalizing right-skewed amount densities).
- **Imbalance Resampling:** Stratified 80/20 train-test splits were established. Synthetic Minority Over-sampling Technique (SMOTE) was calibrated to sample the minority fraud space to a targeted 30% sample target layout, avoiding over-fitting while optimizing precision boundaries.

3. Quantitative Model Comparison

Three distinct architectures representing gradient-boosted trees and distance anomalies were trained and cross-compared against rigorous validation sets. LightGBM significantly outperformed competing models, proving to be the optimal asset for real-time inference tasks due to its asymmetric leaf-wise tree growth optimization.

Model Architecture	Accuracy	Precision	Recall	F1-Score	ROC-AUC	PR-AUC
LightGBM (Baseline Balanced)	0.9796	0.7449	0.6325	0.6841	0.9525	0.7288
XGBoost Classifiers	0.8549	0.1759	0.8541	0.2918	0.9303	0.6221
Isolation Forest (Unsupervised)	0.9533	0.1839	0.0970	0.1270	0.6985	0.0984

Hyperparameter & Threshold Optimization

Automated optimization hyper-tuning was executed via Optuna over 30 targeted trials to maximize Precision-Recall Area Under Curve (PR-AUC), which functions as a superior evaluation metric compared to standard accuracy under steep class imbalances. Post-tuning, the classification decision boundary threshold was dynamically optimized. Sliding the threshold to an **optimal 0.48 cutoff** successfully maximized the F1-score metric to **0.6863**, shifting the operational Recall to **64.72%** with a solid accompanying Precision of **73.05%**.

4. Operational Threat Risk Segmentation

To enable automated workflows within financial fraud units, the output probabilities of the optimized model were mapped to distinct action-oriented Operational Risk Tiers. Below are the verified distribution matrices observed across the validation test space (118,108 transactions):

Risk Tier Category	Probability Threshold	Transaction Volume	Observed Fraud Rate	Prescribed Operational Actions
 Critical Risk	$P(\text{Fraud}) \geq 0.75$	2,186 txns	89.40%	Immediate Hard Block; Auto-Suspend Card Node; Prompt SMS Audit.
 Suspicious Risk	$0.40 \leq P(\text{Fraud}) < 0.75$	2,212 txns	41.10%	Escalate to Manual Verification Queue; Trigger Interactive 2FA.
 Clear / Safe	$P(\text{Fraud}) < 0.40$	113,710 txns	1.10%	Auto-Approve; Pass through payment rails seamlessly.

Dominant Fraud Patterns Identified

- **Night-Hour Anomalies:** Within the Critical Risk tier, 51.8% of malicious activity takes place during dark hours (00:00 to 05:00), contrasting significantly with standard consumer volume.
- **Contextual Magnitude:** Transaction amounts in the upper quartile (\$0.99 standard-scaled value limits) are disproportionately linked to critical alerts.
- **Card Metadata Clusters:** Specific encodings within the `card6` (charge mechanism) and `C14` feature nodes indicate a strong mathematical overlap with compromised merchant profiles.

5. Explainable AI (XAI) & Operational Oversight

To satisfy regulatory transparency guidelines and empower fraud analysts during manual audits, global and local model decisions were mapped using SHAP (SHapley Additive exPlanations) values. The global summary indicated

that charge mechanism structures (`card6`), count velocities (`C14`), and time metrics (`HourOfDay`) are the strongest drivers of risk.

Locally, the application extracts exact contribution weights for single transactions. Positive SHAP values isolate the precise trigger variables pushing a payment toward an alert, while negative SHAP weights reveal safety triggers supporting legitimacy. This technical architecture bridges the gap between high-performing black-box modeling and plain-English operational accountability.

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